For Examiner's Use

9 **(a)** **(i)** Define the *magnetic flux.*

.....

..... [1]

(ii) State the *laws of electromagnetic induction*.

.....

.....

.....

.....

..... [2]

(b) The variation of the magnetic flux density *B* beyond one end of a large bar magnet with distance is shown in Fig. 9.1.

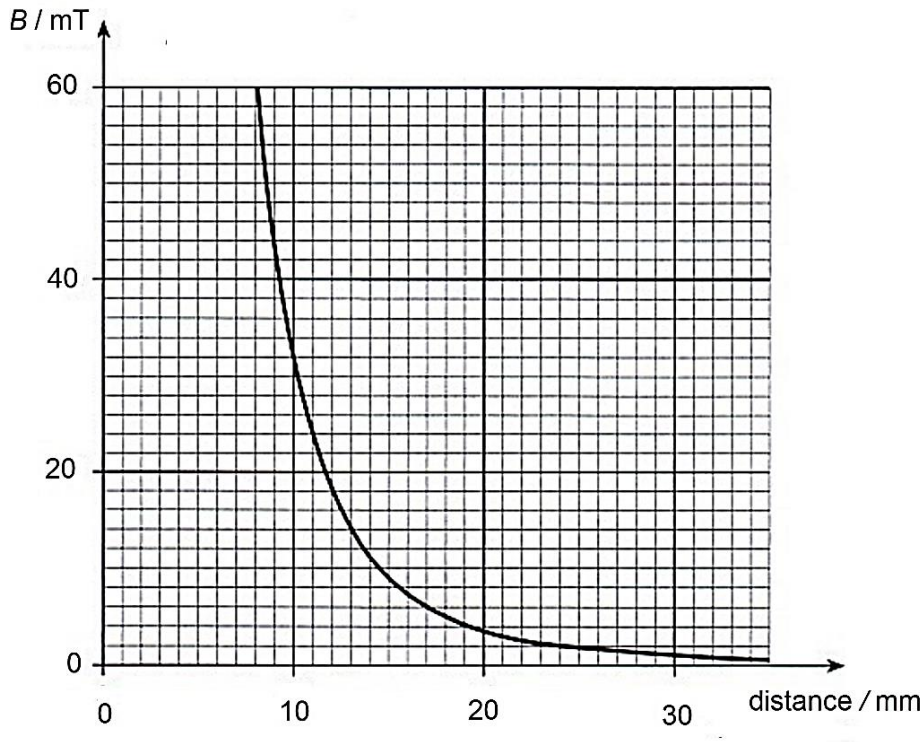

Fig. 9.1

A circular coil of wire with 50 turns and cross-sectional area 16 cm^2 is placed a few centimetres beyond the end of the above stated bar magnet. The axis of the coil is aligned with the axis of the magnet.

(i) Calculate the magnetic flux linkage through the coil when it is 30.0 mm and when it is 10.0 mm from the end of the magnet respectively.

magnetic flux linkage at 30.0 mm = …… Wb

magnetic flux linkage at 10.0 mm = …… Wb [3]

(ii) The coil of wire is moved towards the magnet from the 30.0 mm position to the 10.0 mm position so that a steady e.m.f. of 0.80 mV is induced in it. Calculate the average speed of the movement of the coil.

average speed = …… mm s^{-1} [2]

(iii) Describe how the speed of the coil would need to change when moving from the 30.0 mm position to the 10.0 mm position so as to maintain a steady e.m.f..

……..

……..

……..

……..

…… [2]

For Examiner's Use

(c) An a.c. generator consists of a rectangular coil of 800 turns with the dimensions 5.0 cm × 8.0 cm in a uniform magnetic field of magnitude 0.50 T. The coil has a resistance of 0.60 Ω and it is connected to an external load of resistance 11.4 Ω in a complete circuit. The coil is rotating at a constant speed of 240 revolutions per minute.

(i) Calculate the maximum voltage produced by this generator.

maximum voltage = …….. V [2]

(ii) State the relationship between the induced voltage V and the magnetic flux linkage Φ.

[1]

(iii) Fig. 9.2 shows the variation of the induced voltage V in the coil with time.

Draw on Fig. 9.2, the corresponding variation of the magnetic flux linkage Φ with time.

[2]

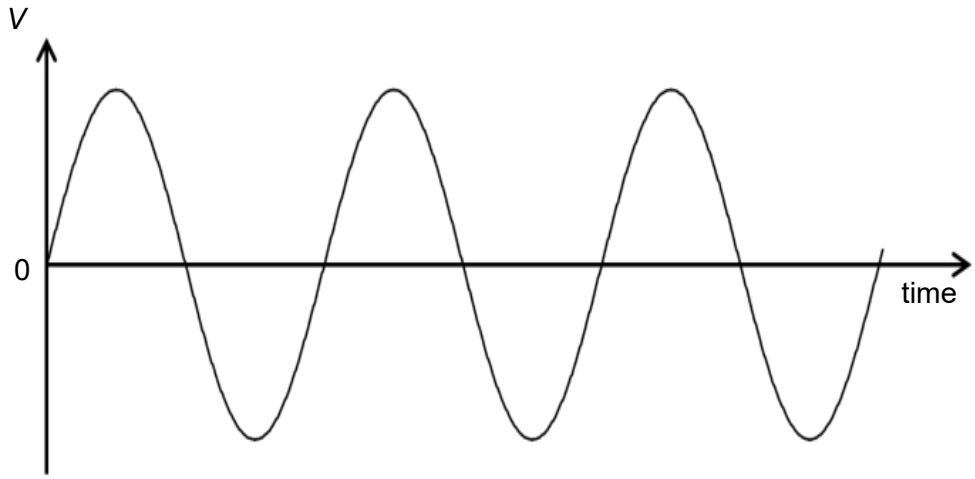

Fig. 9.2

For Examiner's Use

(iv) Calculate the maximum current through the external load.

current = ... A [1]

(v) Explain the terms

(1) *root-mean-square* (r.m.s.) value

.....

.....

(2) r.m.s. *current*.

.....

..... [2]

(vi) Calculate the r.m.s. current through the external load.

r.m.s current = ……. A [1]

(vii) Calculate the average power through the external load.

average power = ……. W [1]

END OF PAPER